

Demetri Pappas

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EDUCATION

University of Central Florida

May 2027

B.S. in Computer Science

GPA 3.6

University of Miami

Transferred 2024

Presidential Scholar (National Merit Finalist), Benacquisto Scholar, Foote Fellow Honors Student.

EMPLOYMENT

Puretalk AI– ML Engineering Intern

May - Sep 2025

Orlando, FL

- Built Python-based evaluation pipelines to benchmark proprietary LLM (RUTH©) performance across reasoning, retrieval, and instruction-following tasks, enabling rapid regression testing between model versions.
- Performed large-scale error analysis across thousands of model outputs, developing failure-mode taxonomies that guided improvements to prompting and retrieval systems.
- Conducted comparative benchmarking of proprietary transformer models and retrieval-augmented generation systems, analyzing tradeoffs between factual accuracy, inference latency, and operational cost.

Puretalk AI– Software Engineering Intern

Jan - May 2025

Remote, NY

- Optimized the platform's landing page and onboarding flow by implementing API-driven interactive components and improved UI/UX.
- Developed five additional dashboards and AI-Agent interfaces with React, Tailwind, and Next.js.
- Implemented Websocket integrations with Azure endpoints to facilitate low-latency AI chatbot, voice agent, and text-to-speech interactions.

FlexPT, LLC– Founder, Fullstack Developer

May 2025 - Current

Orlando, FL

- Built the frontend with React using dynamic UI components and state management for realtime data.
- Designed backend services with Supabase (PostgreSQL), implementing authentication, database operations, and server-side business logic.
- Deployed the application on Vercel, optimizing request routing and helper functions for scale, and deploying serverless functions for low-latency business logic operations.
- Implemented AI workout program generation and editing with the Open AI API, designing custom system prompts and a context-retrieval pipeline to minimize token usage and reduce latency.

CDI Design Group– Frontend Developer

Aug 2024- 2025

- Designed and implemented 10+ responsive, high-fidelity web pages for clients using React, Next.js, and Tailwind CSS.
- Created interactive animated interfaces with Framer Motion, enhancing user engagement following principles of Emotional Design.
- Integrated secure payment flows and subscription management with Stripe API.
- Implemented middleware to protect routes and prevent flashes of unfiltered content.

PROJECTS

CertForge– Microsoft Agents League Hackathon

Jun 2026

- Engineered a typed multi-agent orchestration framework in Python that coordinates project intake, knowledge retrieval, planning, assessment, and validation through modular Pydantic v2 data models and deterministic execution pipelines.
- Implemented retrieval-augmented generation infrastructure using Azure AI Search and local semantic knowledge stores, with fail-safe fallback mechanisms, lazy-loaded cloud dependencies, and managed-identity authentication.
- Containerized and deployed an application on Azure Container Apps with CI/CD-ready infrastructure, structured telemetry logging, RBAC, and Azure OpenAI integration for intelligent workflows.

Transformer Language Model– Pytorch

Jun 2026

- Implemented and trained transformer language models from first principles, reproducing key architectural mechanisms underlying modern large language models.

- Conducted experiments on attention mechanisms and representation learning, analyzing how information propagates through transformer layers during autoregressive generation.
- Developed evaluation and visualization tools for interpreting attention-head specialization, token interactions, and contextual information flow.

High Performance Memory Allocator– C

Mar 2026

- Designed compact metadata headers and pointer arithmetic routines for efficient block traversal and management.
- Optimized allocation and free paths by dynamically reclassifying blocks between small-size buckets and a red-black tree for larger allocations, preserving constant-time list operations and balanced tree invariants during coalescing and splitting.
- Benchmarked performance against 100,000 allocation cycles, evaluating fragmentation and allocation throughout.

SKILLS

ML/AI: PyTorch, Orchestration, LLMs, RAG, LLM Evaluation, Python, EDA, Transformer Language Models.

Software: Full-Stack Development, REST APIs, CI/CD, System Design, Database Design, Git, UI/UX.